

System Design (Telusko) - Day 10

One-Shot 5h course | Video 2:15-2:30

CAP Theorem & Consistency

Understand the trade-off between consistency, availability and partition tolerance.

Key concepts

- CAP: under a network partition you must choose Consistency or Availability.
- CP vs AP systems - pick per use case.
- Consistency models: strong, eventual, read-your-writes.
- Most distributed systems tune for eventual consistency.

Key example

Bank balance -> CP (strong); social likes -> AP (eventual) is fine.

Today's tasks (1.5h)

- ☐ Explain CAP simply
- ☐ Classify 2 systems as CP or AP
- ☐ Define eventual consistency

Resources

Video: youtu.be/Vnm-ycSfJx4 (start at 2:15-2:30)

Docs: docs.telusko.com/docs/system-design

(Full clickable links are in the app: Goals > this day > Study block.)